DIGITAL FORENSICS

Laboratory Report

Lab 4A

BitLocker RAM Key Extraction & Disk Decryption

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Course	Digital Forensics
Lab	4A — Option 1: BitLocker Key Extraction from RAM
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Classification	Academic — Not for Distribution

1. Objective

This laboratory exercise demonstrates that BitLocker full-disk encryption, while effective at protecting data at rest, does not protect against in-memory key extraction when a volume is actively unlocked. The goal was to:

- Boot and unlock a BitLocker-protected Windows 10 virtual machine.
- Acquire a full RAM dump while the encrypted volume remained unlocked.
- Extract the Full Volume Encryption Key (FVEK) from memory using Volatility 3.
- Mount the encrypted disk image using the recovered key via bdemount.
- Recover the embedded challenge flag from the decrypted filesystem.

2. Lab Environment

Host OS	CAINE Linux (forensic live environment)
Guest VM	Windows 10 22H2 (VMware Workstation)
VM Credentials	Username: admin | Password: iloveyou
BitLocker Password	princess
RAM Acquisition	Belkasoft RAM Capturer x64 (WinPmem attempt unsuccessful — Belkasoft RAM Capturer used instead)
Memory Analysis	Volatility 3 with custom BitLocker plugin (vol3_plugins)
Disk Tools	qemu-img, mmls (TSK), dd, xxd
Volume Mounting	bdemount (libbde-utils)
Evidence File	BitLockerUE2RAMDump.img (supplied VMDK)

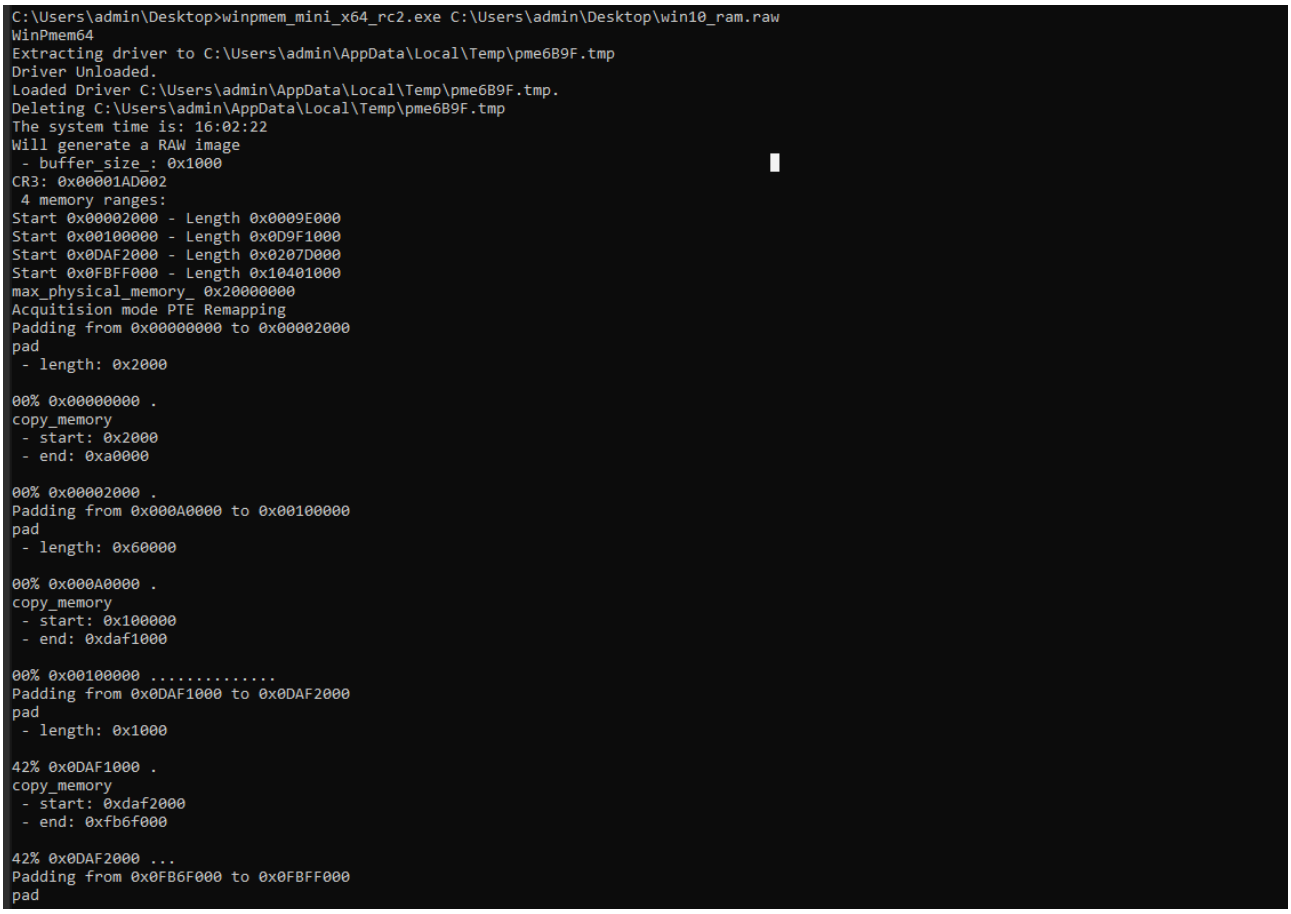
3. Forensic Procedure

3.1 Windows VM Preparation

The provided Windows 10 virtual machine was launched in VMware Workstation and authenticated with the supplied credentials. The BitLocker-protected drive was then unlocked using the passphrase princess. Following successful authentication, the encrypted volume appeared as drive G: containing the file flag.txt, confirming that the FVEK was now loaded into system RAM.

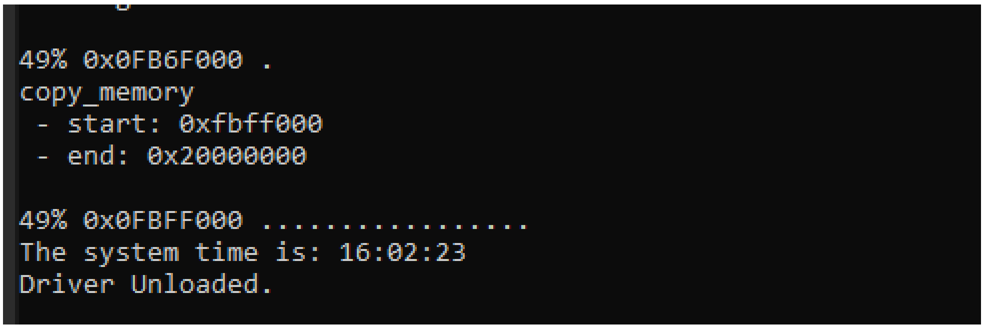
3.2 RAM Acquisition

An initial acquisition attempt using WinPmem was unsuccessful — Volatility could not parse the resulting image. Belkasoft RAM Capturer (x64 edition) was substituted and produced a valid dump of approximately 512 MB, consistent with the VM's configured memory.



```
C:\Users\admin\Desktop>winpmem_mini_x64_rc2.exe C:\Users\admin\Desktop\win10_ram.raw
WinPmem64
Extracting driver to C:\Users\admin\AppData\Local\Temp\pme6B9F.tmp
Driver Unloaded.
Loaded Driver C:\Users\admin\AppData\Local\Temp\pme6B9F.tmp.
Deleting C:\Users\admin\AppData\Local\Temp\pme6B9F.tmp
The system time is: 16:02:22
Will generate a RAW image
 - buffer_size_: 0x1000
CR3: 0x00001AD002
 4 memory ranges:
Start 0x00002000 - Length 0x0009E000
Start 0x00100000 - Length 0x0D9F1000
Start 0x0DAF2000 - Length 0x0207D000
Start 0x0FBFF000 - Length 0x10401000
max_physical_memory_ 0x20000000
Acquitision mode PTE Remapping
Padding from 0x00000000 to 0x00002000
pad
 - length: 0x2000

00% 0x00000000 .
copy_memory
 - start: 0x2000
 - end: 0xa0000

00% 0x00002000 .
Padding from 0x000A0000 to 0x00100000
pad
 - length: 0x60000

00% 0x000A0000 .
copy_memory
 - start: 0x100000
 - end: 0xdaf1000

00% 0x00100000 .............
Padding from 0x0DAF1000 to 0x0DAF2000
pad
 - length: 0x1000

42% 0x0DAF1000 .
copy_memory
 - start: 0xdaf2000
 - end: 0xfb6f000

42% 0x0DAF2000 ...
Padding from 0x0FB6F000 to 0x0FBFF000
pad
```

Figure SC-INTRO-1 — *Initial WinPmem RAM acquisition attempt*



```
49% 0x0FB6F000 .
copy_memory
 - start: 0xfbff000
 - end: 0x20000000

49% 0x0FBFF000 .................
The system time is: 16:02:23
Driver Unloaded.
```

Figure SC-INTRO-2 — *Belkasoft RAM Capturer executing on the Windows VM*

Tool path:

```
C:\Users\admin\Desktop\Belkasoft_RAM_Capture\x64
```

Output file:

```
20260531.mem  (~512 MB)
```

The dump was transferred to the Linux host using the VMware shared folder, as shown in the transfer screenshot:

```
# Linux host — copy RAM dump from VMware shared folder
cp /mnt/hgfs/shared/win10_ram.raw .

# Verify copied RAM dump
ls -lh
```

```
caine@caine-vm:~/lab4/option1_bitlocker$ cp /mnt/hgfs/shared/win10_ram.raw .
caine@caine-vm:~/lab4/option1_bitlocker$ ls -lh
total 512M
-rwxr-xr-x 1 caine caine 512M May 30 18:50 win10_ram.raw
caine@caine-vm:~/lab4/option1_bitlocker$ ls
BitLockerUE2RAMDump.img  win10_ram.raw
caine@caine-vm:~/lab4/option1_bitlocker$
```

Figure SC-INTRO-3 — *RAM dump transfer via VMware shared folder*

3.3 Disk Image Preparation

The supplied BitLocker evidence image (BitLockerUE2RAMDump.img) was identified as a VMDK and converted to raw format before partition analysis:

```
caine@caine-vm:~/bitlocker_lab$ sudo ls -lh
total 1.6G
-rw-rw-r-- 1 caine caine   66 May 31 02:09 0xc58feaba9d20-Dislocker.fvek
-rw-rw-r-- 1 caine caine   66 May 31 02:09 0xc58feabf2c90-Dislocker.fvek
-rw-rw-r-- 1 caine caine   66 May 31 02:09 0xc58feda4bc20-Dislocker.fvek
-rw-rw-r-- 1 caine caine   66 May 31 02:09 0xc58fee492a50-Dislocker.fvek
-rw-rw-r-- 1 caine caine   66 May 31 02:09 0xf8003fa51a50-Dislocker.fvek
dr-xr-xr-x 2 root  root     0 May 31 02:29 bde_mount
-rw-r--r-- 1 caine caine 524M May 31 00:47 bitlocker_disk.raw
-rw-rw-r-- 1 caine caine 506M May 31 00:50 bitlocker_partition.dd
-rw-rw-r-- 1 caine caine  43M Oct 22  2025 BitLockerUE2RAMDump.img
drwxrwxrwx 1 root  root  4.0K Oct 22  2025 clear_mount
drwxrwxr-x 2 caine caine 4.0K May 31 02:24 decrypted_mount
drwxrwxr-x 2 caine caine 4.0K May 31 02:24 dislocker_mount
-rw-rw-r-- 1 caine caine 512M May 31 01:08 ramdump_mmap.raw
-rw-rw-r-- 1 caine caine 512M May 30 23:50 ramdump.raw
drwxrwxr-x 3 caine caine 4.0K May 31 01:30 volatility3-bitlocker
```

Figure SC-01 — *Working directory — RAM dump, disk image, raw disk, extracted partition*

```
# Identify image format
qemu-img info BitLockerUE2RAMDump.img
```

```
# Convert VMDK to raw
qemu-img convert -O raw BitLockerUE2RAMDump.img bitlocker_disk.raw
```

```
# Inspect partition table
mmls bitlocker_disk.raw
```

```
# Extract BitLocker data partition
dd if=bitlocker_disk.raw of=bitlocker_partition.dd \
   bs=512 skip=32768 count=1036288 status=progress
```

```
# Verify BitLocker signature
xxd -g 1 -l 16 bitlocker_partition.dd
```

```
caine@caine-vm:~/bitlocker_lab$ qemu-img info BitLockerUE2RAMDump.img
image: BitLockerUE2RAMDump.img
file format: vmdk
virtual size: 524M (549453824 bytes)
disk size: 43M
cluster_size: 65536
Format specific information:
    cid: 985174899
    parent cid: 4294967295
    create type: monolithicSparse
    extents:
        [0]:
            virtual size: 549453824
            filename: BitLockerUE2RAMDump.img
            cluster size: 65536
            format:
```

Figure SC-02 — *qemu-img info — VMDK identified, 524 MB virtual size*

```
caine@caine-vm:~/bitlocker_lab$ mmls bitlocker_disk.raw
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  -------   0000000000   0000000033   0000000034   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000000034   0000032767   0000032734   Microsoft reserved partition
005:  001       0000032768   0001069055   0001036288   Basic data partition
006:  -------   0001069056   0001073151   0000004096   Unallocated
caine@caine-vm:~/bitlocker_lab$
```

Figure SC-03 — *mmls partition table — start sector 32768, length 1,036,288 sectors*

```
caine@caine-vm:~/bitlocker_lab$ ls -lh bitlocker_partition.dd
xx-rw-rw-r-- 1 caine caine 506M May 31 00:50 bitlocker_partition.dd
caine@caine-vm:~/bitlocker_lab$ xxd -g 1 -l 16 bitlocker_partition.dd
00000000: eb 58 90 2d 46 56 45 2d 46 53 2d 00 02 08 00 00  .X.-FVE-FS-.....
caine@caine-vm:~/bitlocker_lab$
```

Figure SC-04 — *xxd output — -FVE-FS- BitLocker signature confirmed*

The mmls output identified the BitLocker data partition at:

Start Sector	32768
Length (sectors)	1,036,288
Signature Verified	Yes — -FVE-FS- present at offset 0x03

3.4 FVEK Extraction with Volatility 3

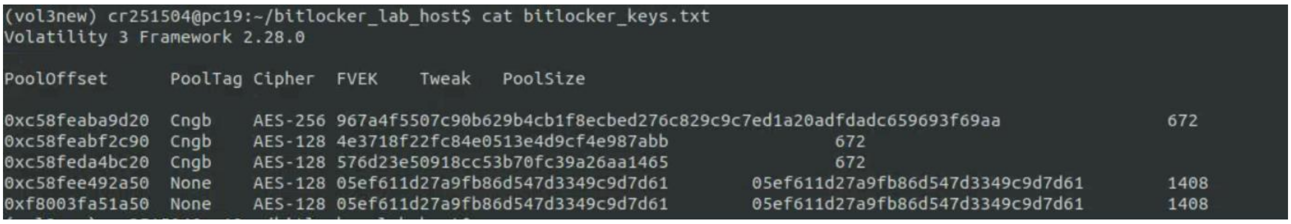
Volatility 2 profiles bundled with CAINE did not support the Windows 10 22H2 memory layout. Volatility 3, running on the Linux host with a custom BitLocker plugin, successfully parsed the Belkasoft dump.

```
# Confirm memory image identity
vol -f belka_ram.raw windows.info.Info

# Scan for BitLocker FVEK candidates
vol -p ./vol3_plugins -f belka_ram.raw \
    windows.bitlocker.BitlockerFVEKScan \
```

```
    --tags FVEc Cngb None dFVE \
    --dislocker | tee bitlocker_keys.txt
```

```
cr251504@pc19:~/bitlocker_lab_host$ cd ~/bitlocker_lab_host
source vol3new/bin/activate
vol -f belka_ram.mem windows.info.Info
Volatility 3 Framework 2.28.0
Progress:  100.00               PDB scanning finished
Variable        Value

Kernel Base     0xf8003e601000
DTB     0x1ad000
Symbols file:///home/fluttershy-01/cr251504/bitlocker_lab_host/vol3new/lib/python3.10/site-packages/volatility3/symbols/windows/ntkrnlmp.pdb/68A17FAF3012B7846079AEECDBE0
A583-1.json.xz
Is64Bit True
IsPAE   False
layer_name      0 WindowsIntel32e
memory_layer    1 FileLayer
KdVersionBlock  0xf8003f210398
Major/Minor     15.19041
MachineType     34404
KeNumberProcessors      4
SystemTime      2026-05-30 23:57:44+00:00
NtSystemRoot    C:\Windows
NtProductType   NtProductWinNt
NtMajorVersion  10
NtMinorVersion  0
PE MajorOperatingSystemVersion  10
PE MinorOperatingSystemVersion  0
PE Machine      34404
PE TimeDateStamp        Wed Jun 28 04:14:26 1995
(vol3new) cr251504@pc19:~/bitlocker_lab_host$
```

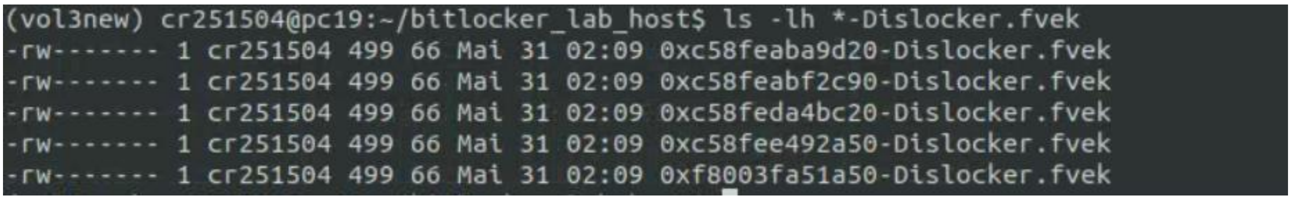
Figure SC-05 — *Volatility 3 windows.info.Info — Windows 10 x64 memory image confirmed*



```
(vol3new) cr251504@pc19:~/bitlocker_lab_host$ cat bitlocker_keys.txt
Volatility 3 Framework 2.28.0

PoolOffset      PoolTag Cipher  FVEK    Tweak   PoolSize

0xc58feaba9d20  Cngb    AES-256 967a4f5507c90b629b4cb1f8ecbed276c829c9c7ed1a20adfdadc659693f69aa                672
0xc58feabf2c90  Cngb    AES-128 4e3718f22fc84e0513e4d9cf4e987abb                672
0xc58feda4bc20  Cngb    AES-128 576d23e50918cc53b70fc39a26aa1465                672
0xc58fee492a50  None    AES-128 05ef611d27a9fb86d547d3349c9d7d61        05ef611d27a9fb86d547d3349c9d7d61        1408
0xf8003fa51a50  None    AES-128 05ef611d27a9fb86d547d3349c9d7d61        05ef611d27a9fb86d547d3349c9d7d61        1408
```

Figure SC-06 — *Volatility 3 BitlockerFVEKScan — FVEK/TWEAK candidates listed*

The plugin located multiple FVEK/TWEAK candidates and wrote Dislocker-compatible .fvek files to disk. The key pair that successfully decrypted the volume was:



```
(vol3new) cr251504@pc19:~/bitlocker_lab_host$ ls -lh *-Dislocker.fvek
-rw------- 1 cr251504 499 66 Mai 31 02:09 0xc58feaba9d20-Dislocker.fvek
-rw------- 1 cr251504 499 66 Mai 31 02:09 0xc58feabf2c90-Dislocker.fvek
-rw------- 1 cr251504 499 66 Mai 31 02:09 0xc58feda4bc20-Dislocker.fvek
-rw------- 1 cr251504 499 66 Mai 31 02:09 0xc58fee492a50-Dislocker.fvek
-rw------- 1 cr251504 499 66 Mai 31 02:09 0xf8003fa51a50-Dislocker.fvek
```

Figure SC-07 — *Dislocker-compatible .fvek files generated by the Volatility plugin*

FVEK	05ef611d27a9fb86d547d3349c9d7d61
TWEAK	05ef611d27a9fb86d547d3349c9d7d61

3.5 BitLocker Volume Mounting and Flag Recovery

The extracted partition was mounted using bdemount with the recovered key pair, then mounted as a read-only loop device:

```
# Mount BitLocker partition with recovered FVEK
sudo bdemount -k 05ef611d27a9fb86d547d3349c9d7d61:05ef611d27a9fb86d547d3349c9d7d61 \
     bitlocker_partition.dd bde_mount/

# Mount decrypted volume read-only
sudo mount -o loop,ro bde_mount/bde1 clear_mount/

# List filesystem contents
ls -la clear_mount/

# Read the flag
cat clear_mount/flag.txt.txt
```

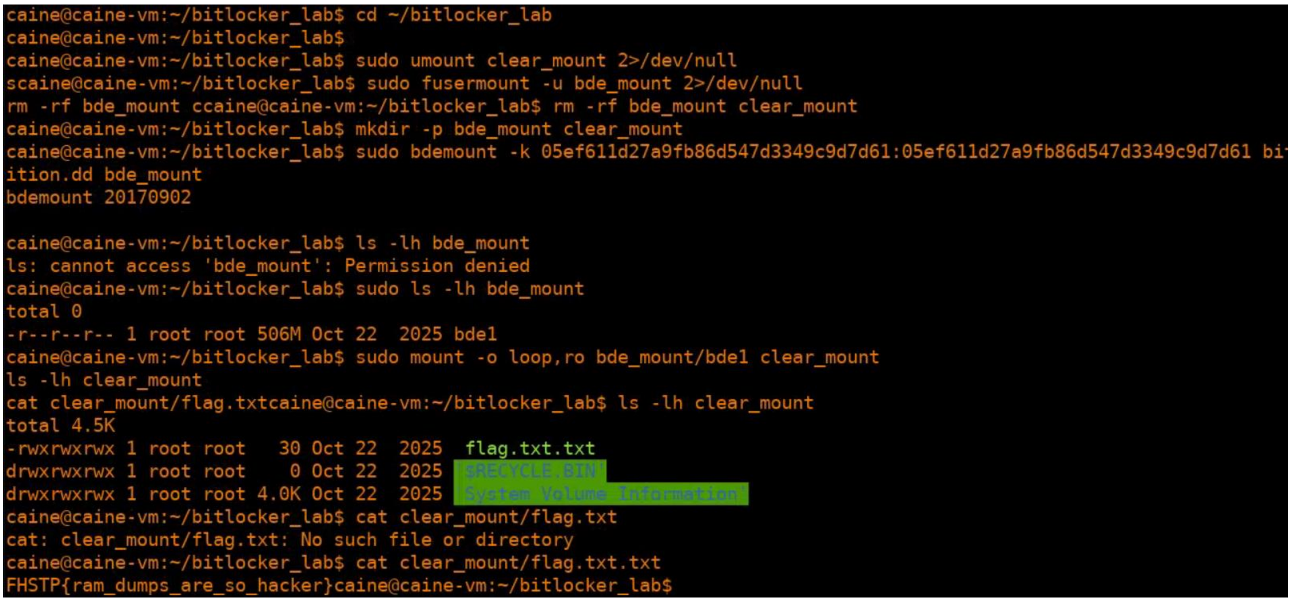


Figure SC-08 — *bdemount + mount + cat flag.txt.txt — flag recovered*

The decrypted filesystem contained the following artifacts:

- flag.txt.txt
- $RECYCLE.BIN
- System Volume Information

4. Recovered Flag

RECOVERED FLAG

`FHSTP{ram_dumps_are_so_hacker}`

The flag was recovered from flag.txt.txt on the decrypted BitLocker volume. Its presence confirms complete and successful execution of all forensic steps — from RAM acquisition through key extraction to volume decryption.

5. Key Findings & Conclusion

This laboratory confirmed the following forensic findings:

- In this attempt, the WinPmem memory image could not be parsed correctly by Volatility, so Belkasoft RAM Capturer was used instead.
- Volatility 2 profiles in CAINE are insufficient for modern Windows 10 builds. Volatility 3 with a current symbol set is required.
- BitLocker FVEKs reside in cleartext in RAM while a volume is unlocked and are recoverable through memory forensics.
- bdemount successfully decrypted the BitLocker partition using the extracted FVEK/TWEAK pair without requiring the original passphrase.

From a defensive standpoint, this exercise underscores the importance of hibernation and memory protections (e.g., Secure Boot, Kernel DMA Protection) in environments where physical or remote memory access is a realistic threat vector. BitLocker alone does not protect against cold-boot or live-memory attacks.

6. Screenshot Evidence Index

The following screenshots were captured during the lab to support the documented findings. Each figure appears inline within the relevant procedural section above.

Reference	Description	Forensic Relevance
SC-INTRO-1	Initial WinPmem RAM acquisition attempt	Documents the first RAM acquisition attempt before Belkasoft RAM Capturer was used
SC-INTRO-2	Belkasoft RAM Capturer executing on the Windows VM	Documents RAM acquisition tool and execution
SC-INTRO-3	RAM dump transfer via VMware shared folder	Shows the file transfer method and resulting 512 MB RAM dump on the Linux host
SC-01	Working directory — RAM dump, disk image, raw disk, extracted partition	Establishes chain of custody for all evidence files
SC-02	qemu-img info output — encrypted VMDK (524 MB virtual size)	Confirms image format and integrity before conversion
SC-03	mmls partition table — start sector 32768, length 1,036,288 sectors	Identifies correct partition boundaries for extraction
SC-04	xxd output showing -FVE-FS- BitLocker signature	Verifies extracted partition is a valid BitLocker volume
SC-05	Volatility 3 windows.info.Info — Windows 10 x64 memory layout	Validates RAM dump parsability and OS identification
SC-06	Volatility 3 BitlockerFVEKScan — FVEK/TWEAK candidates listed	Primary evidence of key extraction from memory
SC-07	Dislocker-compatible .fvek files generated by Volatility plugin	Intermediate artefacts linking memory scan to mount step
SC-08	bdemount + mount + cat flag.txt.txt — flag recovered	Confirms successful decryption and flag recovery